Breast Cancer Classification Using Probabilistic Machine Learning: Logistic Regression versus Gaussian Naive Bayes

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***Abstract*—In the task of breast tumor classification based on digitized biopsy data, probability estimates are utilized for carrying out classification. In this study, logistic regression (LR) is compared to Gaussian Naive Bayes (GNB), the latter of which assumes the conditional independence of the features under consideration in the process of classification. The dataset used in the study is the Wisconsin Diagnostic Breast Cancer dataset, which is composed of 569 records; the data included thirty features based on ten characteristics of cell nuclei. The data was split into 455 training and 114 test records; the standardization process was applied only to the training data. In the end, logistic regression achieved an accuracy rate of 98.25 percent and a ROC-AUC value of 0.9954, which is better than the results of GNB (whose accuracy was 92.98 percent and whose ROC-AUC value was 0.9868).**

***Index Terms*—breast cancer classification, logistic regression, maximum likelihood estimation, Gaussian naive Bayes, Bayesian inference, ROC analysis.**

I. Introduction

Detecting breast cancer requires precise findings regarding tumors. The "Wisconsin Diagnostic Breast Cancer (WDBC)" dataset contains numerical details of cell nuclei. These features are used to classify tumors based on supervised methods [1], [2]. Although computer-assisted techniques aid in providing reliable assessments, they are not a substitute for clinical procedures.

This study compares logistic regression (LR) and Gaussian Naive Bayes (GNB) for the practical classification of breast cancers based on conditional probabilities. LR calculates the probability of classes using the method of maximum likelihood and regularization [3], [7], [8]. In contrast, GNB estimates class priors and makes use of Bayes' theorem for posterior estimates [4], [6].

The evaluation of these techniques was performed using test data and five-fold cross-validation. The evaluation also examines the assumptions of the models, the processes of probability estimation and the coefficients of LR, and it identifies the errors made by each of the two methods.

II. Problem Statement and Related Work

A. Problem Statement

Let $x \in R^{30}$ be the 30 measurements obtained from a biopsy image and $y \in \{0, 1\}$ be the diagnosis, where $y = 0$ refers to malignant and $y = 1$ to benign in the given experiment. The goal was to estimate $P(y \mid x)$ and predict a class using a threshold. The techniques being compared here were logistic regression and Gaussian Naive Bayes. The evaluation covered accuracy, precision, recall, specificity, F1 score, ROC-AUC, and five-fold cross-validation results. The additional goal was to determine which features are related to the LR predictions and to determine when the assumptions of either model are violated.

B. Related Work

According to Street, Wolberg, and Mangasarian, cell nuclei features can be useful in differentiating between malignant and benign breast tumors, forming the basis for the WDBC dataset [2], which can be found in the UCI Machine Learning Repository [1].

Ng and Jordan studied logistic regression (LR) and naive Bayes, focusing on the differences between their approaches [4]. While LR learns the probability of the outcome directly, naive Bayes relies on the prior of the class and the distribution of its features. Some studies have covered the topics of Bayesian inference, regularization, and the interpretation of LR coefficients [6]–[8].

Different machine-learning techniques have been previously applied to the WDBC dataset, such as multi-layer perceptrons, naive Bayes, support vector machines, and LR, with good results [9]–[11]. However, the performance varies with respect to the data preprocessing, model parameters, and evaluation approach used in each study.

Following this line of research, the authors of this paper compared LR and GNB using a stratified train–test split, five-fold cross-validation, ROC-AUC and confusion matrices, in addition to coefficients and predicted probabilities, without claiming any clinical usefulness or general superiority.

III. Dataset and Preprocessing

A. Dataset

The scikit-learn library provides access to the WDBC dataset, which consists of 569 tumor samples, each with 30 features represented by real values. The dataset consists of 212 malignant cases and 357 benign cases with no missing values. Therefore, 37.3% of the total data consist of malignant samples, while the remaining 62.7% are benign. The dataset contains 10 main nuclear features, namely radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension; each feature has three values, namely mean, standard error, and worst value, according to [1], [2]. Table I summarizes the dataset.

TABLE I. Dataset Summary

Property	Value
Dataset	Wisconsin Diagnostic Breast Cancer (WDBC)

B.Tech., Vellore Institute of Technology, Vellore, India

Course: Probability and Statistics (BAMAT207) | Faculty: Dr. Pallavi Mishra

Property	Value
Number of samples	569
Number of features	30 numeric features
Class counts	212 malignant; 357 benign
Missing values	0
Training/test split	455 training; 114 test
Training class counts	170 malignant; 285 benign

B. Split and Standardization

A stratified train-test split of 80/20 was conducted using a random_state value of 42. Stratification ensures that the class proportions are similar in the training and test parts. The training data comprise 455 samples, of which 170 are malignant and 285 are benign. The test data comprise 114 samples. The standardization method was applied to the data using a StandardScaler. The scaler was fit only to the training data and then used on the test set. If we were to apply the preprocessing method to the full dataset, information from the test set would have been used in the training pipeline, resulting in data leakage and reduced independence of the test performance.

The standardization process brings all variables to the same scale, which is important for logistic regression because the size of the coefficients is affected by the scale of the variables. The models used in this experiment received the same scaled inputs. In the five-fold cross-validation, scaling was performed within each fold separately, allowing the models to be tested independently.

C. Exploratory Data Analysis

The exploratory analysis considered several aspects: the balance of the classes, histograms with kernel density estimation, boxplots for each class, and a correlation heatmap (Figs. 1–4). The distribution of the classes is moderately imbalanced in the direction of benign instances, but both classes have sufficient instances to perform stratified testing. The mean values of texture, smoothness, and symmetry reveal a normally distributed pattern with mild right tails. The mean value for the radius shows a right-skewed behavior, indicating that the values are not always Gaussian-distributed. From the class-specific plots, it can be seen that malignant tumors tend to have a larger radius; also, the correlation analysis confirms that the radius, perimeter, and area are extremely strongly correlated with each other (r = 0.99). Such redundancy is very important for the interpretation of the results of Gaussian Naive Bayes, because the assumption of conditional independence means that the algorithm treats each feature likelihood as an independent contribution. Therefore, correlated measurements can provide repeated evidence of approximately the same characteristic of size.

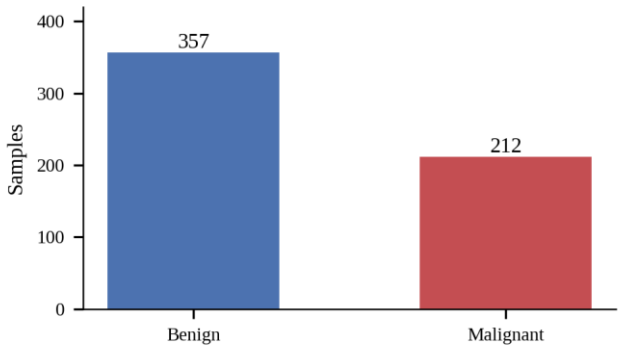


Fig. 1. Class distribution of the 569 WDBC samples (357 benign, 212 malignant).

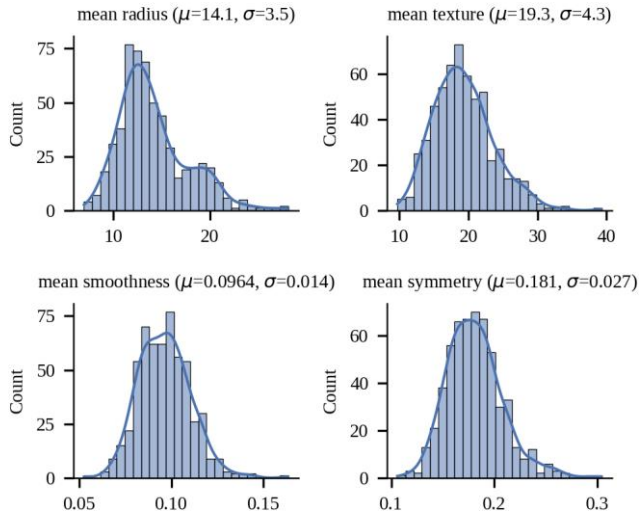


Fig. 2. Histograms with kernel density estimates for four mean-value features; μ and σ are the sample mean and standard deviation.

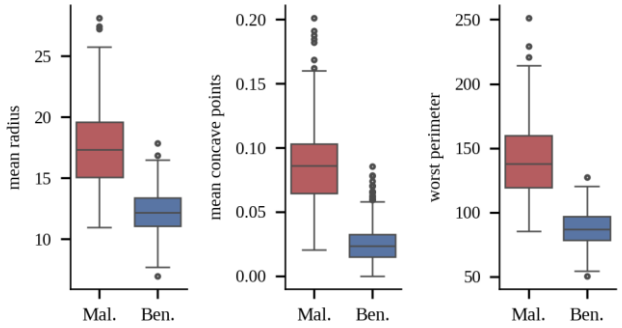


Fig. 3. Boxplots of three features by diagnosis (Mal. = malignant, Ben. = benign).

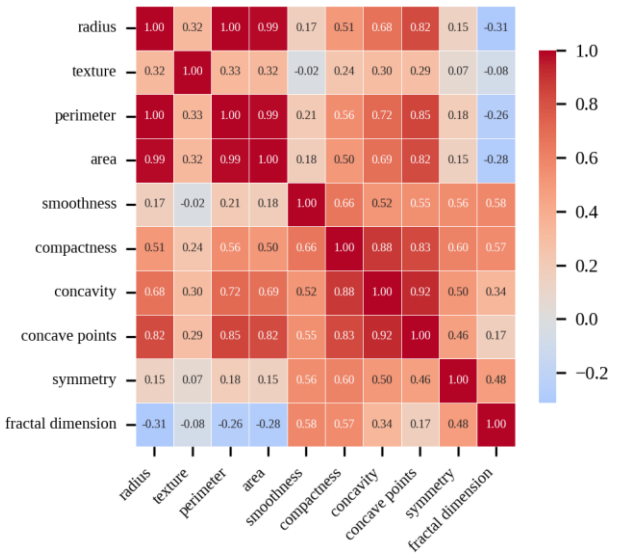


Fig. 4. Pearson correlation matrix of the ten mean-value features.

IV. Mathematical Formulation

A. Conditional Probability and Evaluation Metrics

Conditional probability is a measure of the probability of a certain event given the occurrence of another event. For two events A and B with P(B) > 0,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{1}$$

Model performance is defined via accuracy, precision, recall, specificity, and F1-score. In our research, benign tumors are treated as the positive class when calculating precision, recall and F1-score. Accordingly, TP and TN are benign and malignant samples classified correctly, FP is a malignant sample predicted as benign, and FN is a benign sample predicted as malignant.

Accuracy is the proportion of all predictions that are correct:

$$\text{Accuracy} = \frac{\text{TP+TN}}{\text{TP+TN+FP+FN}} \tag{2}$$

Precision is the proportion of benign predictions that are correct:

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \tag{3}$$

Recall is the proportion of benign cases that are predicted correctly:

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{4}$$

Specificity is the proportion of malignant cases that are identified correctly:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN+FP}} \tag{5}$$

F1-score is the harmonic mean of precision and recall:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision+Recall}} \tag{6}$$

Since benign tumors make up 62.7% of the dataset, accuracy alone cannot show the true performance of the model. Confusion matrices and class-related metrics reveal the misclassifications that occurred within the model, especially malignant tumors misclassified as benign, which can lead to serious implications in practical medicine.

B. Logistic Regression and Maximum Likelihood

The logistic regression method is used to bring the 30 standardized features together into a single score before converting this score into a probability through the use of the sigmoid function. The formula for the linear score is given as follows:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_{30} x_{30} \tag{7}$$

The sigmoid function's equation is as follows:

$$p = \frac{1}{1+\exp(-z)} \tag{8}$$

Here, p = P(benign | x). The positive scores give preference to benign predictions, while negative scores are responsible for malignant predictions. This score can be referred to as the log-odds, which can be expressed mathematically as z = log[p / (1 − p)]. The coefficients for the model are obtained by maximizing the Bernoulli log-likelihood:

$$\ell(\beta) = \sum_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \tag{9}$$

There is no closed-form solution for this equation, so numerical optimization is used. The implemented model used L2 regularization with C = 1.0 and a maximum of 5,000 iterations. The estimated model converged after 19 iterations and has an intercept of 0.3022 [3], [7], [8].

C. Gaussian Naive Bayes and Bayesian Updating

Bayes' theorem states that P(C | x) = P(x | C)P(C) / P(x). Here, P(C) is the prior probability of a class, P(x | C) is the likelihood of x given class C, and P(C | x) is the posterior probability of class C given x. In Gaussian Naive Bayes, each feature of the class is described using a Gaussian distribution with a specific mean and variance for each class. It assumes that features are conditionally independent. Thus, $P(x \mid C) \approx \prod_j P(x_j \mid C)$. The class with the highest posterior probability is chosen.

The training prior for malignancy is P(malignant) = 0.3736. The posterior combines the prior with evidence from the measured features and is normalized over the classes. The assumption of conditional independence of features does not hold well for WDBC, since the radius, perimeter, and area values are correlated. This means that some relevant evidence can be counted multiple times, which can explain the very extreme posterior values and the weaker test performance.

D. ROC Curve and AUC

As the decision threshold varies, the true-positive rate is plotted against the false-positive rate in the receiver operating characteristic (ROC) curve. The ROC-AUC illustrates the ranking performance over various thresholds. In probabilistic terms, AUC can be defined as the probability that a randomly chosen positive example receives a higher score than a randomly chosen example of the other class, with ties handled in the usual way. An AUC of 0.5 refers to a random ranking, while 1.0 refers to an ideal ranking. However, a high AUC does not necessarily guarantee appropriate probability calibration or practical usefulness.

V. Experimental Setup

The experiment was carried out on Google Colab, making use of Python and several libraries (pandas, NumPy, scikit-learn, matplotlib, and seaborn). The WDBC dataset was obtained using scikit-learn. After completing exploratory data analysis, an 80-20 stratified split (random_state = 42) was carried out. Only the training data was used to fit the StandardScaler. Logistic Regression used L2 regularization (C = 1.0 and max_iter = 5000). Gaussian Naive Bayes relied on the default variance smoothing parameter provided in scikit-learn. Both models were evaluated on the same test set. The five-fold cross-validation was done in such a way that scaling was performed within each cross-validation fold. The threshold was set to 0.5, and no hyperparameter search or threshold tuning was carried out.

VI. Results

A. *Test-Set Performance*

The results of the assessment measures of both models are provided in Table II, and the corresponding ROC curves are shown in Fig. 5. Logistic Regression proved to be superior to Gaussian Naive Bayes across all quantities. The rate of accurate predictions of the model was 0.9825 (98.25%) against the value of 0.9298 (92.98%) demonstrated by GNB, which is a difference of 5.27 percentage points.

TABLE II. Test-Set Performance of the Two Models

Metric	Logistic Regression	Gaussian Naive Bayes
Accuracy	0.9825	0.9298
Precision (benign positive)	0.9861	0.9444
Recall (benign positive)	0.9861	0.9444
Specificity (malignant correctly identified)	0.9762	0.9048
F1 score	0.9861	0.9444
ROC-AUC	0.9954	0.9868
5-fold CV AUC (mean ± SD)	0.9952 ± 0.0040	0.9855 ± 0.0103

Using five-fold cross-validation, we observe that the AUC values are also consistent with those obtained through the held-out approach. As observed, the mean AUC for LR is 0.9952 with a standard deviation of 0.0040, whereas for GNB, the mean AUC is 0.9855 with a standard deviation of 0.0103. The lower AUC dispersion across the folds for LR indicates more consistent AUC measurement across the folds. Nevertheless, five folds alone are not sufficient to confirm this in other populations.

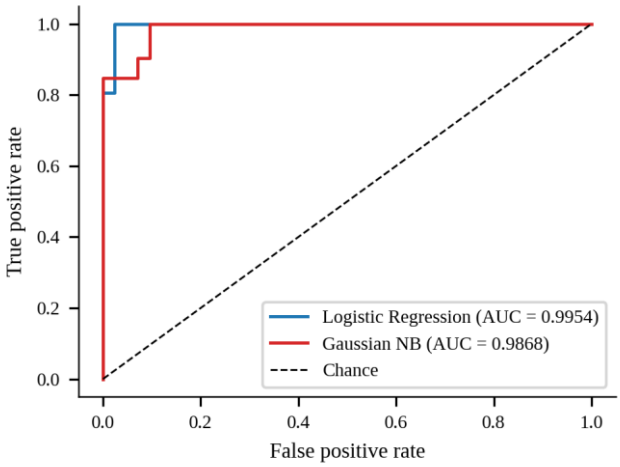


Fig. 5. ROC curves of both models on the test set.

B. *Confusion-Matrix Analysis*

The confusion-matrix counts shown in Table III and Fig. 6 follow the definitions of TN and FP given in Section IV-A. In this case, an FP means that a malignant sample has been predicted as benign. This definition is important to emphasize, since it differs from the one in which malignancy is taken as the positive category.

TABLE III. Confusion-Matrix Counts on the Test Set

Model	TN	FP (malignant predicted benign)	FN (benign predicted malignant)	TP
Logistic Regression	41	1	1	71
Gaussian Naive Bayes	38	4	4	68

In the case of Logistic Regression, accurate identification was achieved in 71 of the 72 benign cases and in 41 of the 42 malignant cases. Consequently, it missed one malignant sample and obtained an effectiveness of 97.6% in identifying malignant cases. On the contrary, Gaussian Naive Bayes correctly classified 68 of 72 benign cases and 38 of 42 malignant cases. Hence, it missed four malignant samples and achieved a success rate of 90.5% in identifying malignant cases. It must be noted that the small number of malignant cases (42) in the test sample makes the results easy to over-interpret; even though there is a notable difference between one and four misclassified malignant cases, it should be analyzed carefully.

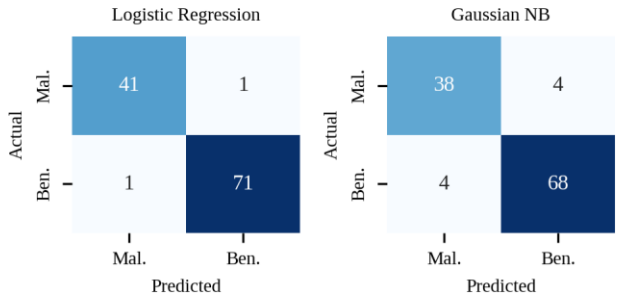


Fig. 6. Confusion matrices on the 114-sample test set (Mal. = malignant, Ben. = benign).

C. Coefficient Interpretation

The LR coefficients apply to standardized features, meaning each coefficient gives the change in the log-odds of the benign class for a one-standard-deviation increase in that feature, while keeping the other features constant. Exponentiating a coefficient gives the odds ratio for the benign class. The largest coefficients are listed in Table IV and plotted in Fig. 7. The most negative coefficients signify a higher probability of malignancy.

TABLE IV. LARGEST STANDARDIZED LOGISTIC REGRESSION COEFFICIENTS

Feature	β	eβ (benign odds ratio)	Malignancy-odds multiplier (approx.)
Worst texture	−1.2551	0.285	3.51
Radius error	−1.0830	0.339	2.95
Worst concave points	−0.9537	0.385	2.60
Worst area	−0.9478	0.388	2.58
Worst radius	−0.9476	0.388	2.58

For instance, if there is a one-standard-deviation increase in the worst texture, the modeled benign odds are multiplied by 0.285 or, equivalently, the malignancy odds are multiplied by 1/0.285 = 3.51 with other predictors constant. Note that this is a conditional model interpretation rather than a causal one. The positive coefficient reported for mean compactness may be a result of multicollinearity, since mean compactness is highly correlated with mean concavity at r = 0.88. Given that the predictors are correlated, it is inappropriate to interpret a single coefficient without considering the other coefficients in the model; the multicollinearity explanation is therefore plausible but unproven.

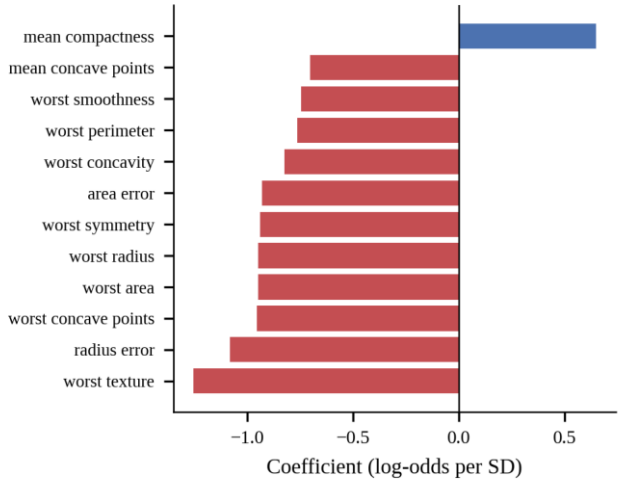


Fig. 7. Largest logistic regression coefficients (standardized features); negative values raise the odds of malignancy.

D. Bayesian Posterior Behavior

Fig. 8 compares the malignancy prior of approximately 0.374 with the posterior probabilities for five test samples. As can be seen in the illustration, the posteriors shown are mostly either below 0.025 or above 0.975, and only one benign sample showed a posterior malignancy probability of 0.098, which raises suspicion about the reliability of the posterior probabilities. One potential cause may be that the Gaussian Naive Bayes model treats correlated measurements as independent samples, and thus makes repeated observations about related information. Note that this figure is simply for illustration and does not demonstrate proper calibration of this model across the entire dataset.

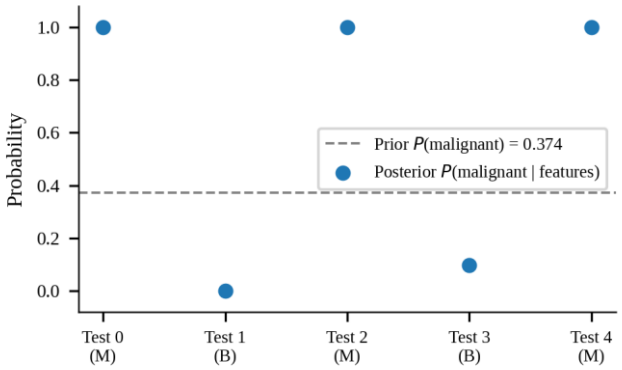


Fig. 8. Prior and posterior probability of malignancy for five test samples (M = malignant, B = benign).

VII. DISCUSSION

Logistic Regression is superior to Gaussian Naive Bayes according to the identified metrics of the testing phase and the AUC from cross-validation. This can be attributed to the underlying distinction between discriminative and generative modeling. Logistic regression is a discriminative model that estimates P(y | x) directly and does not require the features to be independent. Gaussian Naive Bayes, on the other hand, works like a generative model that creates class-conditional distributions and then combines them while assuming

independence. The dataset has several high correlations between features such as radius, perimeter, and area, making the independence assumption unsuitable. This can distort both the classification and the predicted probabilities.

The outcomes of this study agree with those of Hossin et al., who also found LR to be the most effective technique compared with other methods. The current findings are in contrast to those of Soltanpour Gharibdousti et al., who reported that Naive Bayes and SVM can work well after feature reduction. Nonetheless, the results are not contradictory, because the differences can arise from discrepancies in the data preprocessing methods applied, the choice of splits, and the parameter settings of the models used in the studies. It should be mentioned that the findings must be considered a comparison within the defined pipeline, rather than evidence in favor of LR being superior for breast cancer detection. The test set comprises 114 samples (42 of them malignant). One versus four missed malignant cases is a notable difference; however, the study does not conduct a paired statistical test to determine whether the errors of the models differ or not. The dataset is provided by a single well-known source, and no external validation set is applied.

Another limitation is that the hyperparameters chosen for the study were standard/default ones, a threshold of 0.5 was utilized, and there was no calibration evaluation. If the model is to be used mainly for screening, this is likely to require a threshold that favors malignant recall, though changes in the threshold would have an effect on the false positive rates and should be made transparently. A further limitation is the presence of correlated features, which complicates the interpretation of the logistic regression coefficients and goes against the independence assumption of Gaussian Naive Bayes. A thorough analysis should use repeated or nested validation where suitable, and any outcome assessment should use multiple error analyses, calibration plots, and independent datasets.

VIII. Conclusion and Future Work

This study compared Logistic Regression and Gaussian Naive Bayes for the probabilistic classification of the WDBC dataset. Using a stratified 80-20 split and five-fold cross-validation, Logistic Regression achieved an accuracy of 98.25% and an ROC-AUC of 0.9954, versus 92.98% accuracy and 0.9868 for Gaussian Naive Bayes. From the confusion matrices presented, Logistic Regression wrongly predicted one malignant sample as benign, while Gaussian Naive Bayes wrongly predicted four malignant samples as benign. The study illustrates the relationship between conditional probability, maximum likelihood, log-odds, Gaussian likelihoods, Bayesian updating, and ROC-AUC in practical classification results; however, it is not possible to draw conclusions regarding the clinical importance of these methods, as the test set was small, the features were highly correlated, and the threshold was fixed.

Future studies should utilize McNemar's test to analyze paired predictions, find optimal values of regularization and thresholds in terms of malignant recall, implement a calibration plot, compare another model (for example, Random Forest), study the effect of features with SHAP, and provide code along with a README in order to ensure reproducibility.

References

[1] W. Wolberg, O. Mangasarian, N. Street, and W. Street, "Breast Cancer Wisconsin (Diagnostic) [Dataset]," UCI Machine Learning Repository, 1995, doi: 10.24432/C5DW2B.

[2] W. N. Street, W. H. Wolberg, and O. L. Mangasarian, "Nuclear feature extraction for breast tumor diagnosis," in *Proc. SPIE*, vol. 1905, 1993, pp. 861–870, doi: 10.1117/12.148698.

[3] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[4] A. Y. Ng and M. I. Jordan, "On discriminative vs. generative classifiers: A comparison of logistic regression and naive Bayes," in *Advances in Neural Information Processing Systems 14*, 2002, pp. 841–848.

[5] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.

[6] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[7] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York, NY, USA: Springer, 2009.

[8] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Hoboken, NJ, USA: Wiley, 2013.

[9] M. M. Hossin, F. M. J. M. Shamrat, M. R. Bhuiyan, R. A. Hira, T. Khan, and S. Molla, "Breast cancer detection: An effective comparison of different machine learning algorithms on the Wisconsin dataset," *Bulletin of Electrical Engineering and Informatics*, vol. 12, no. 4, 2023, doi: 10.11591/eei.v12i4.4448.

[10] M. Soltanpour Gharibdousti, S. M. Haider, D. Ouedraogo, and S. Lu, "Breast cancer diagnosis using feature extraction techniques with supervised and unsupervised classification algorithms," *Applied Medical Informatics*, vol. 41, no. 1, pp. 40–52, Mar. 2019.

[11] A. F. Agarap, "On breast cancer detection: An application of machine learning algorithms on the Wisconsin diagnostic dataset," arXiv:1711.07831, 2017.